

# Problem Statement

*LLM assistants forget user preferences stated earlier in a conversation (PrefEval, ICLR 2025): accuracy collapses as unrelated context grows. Project: a trained preference-detection classifier + preference memory + modern retrieval. Mid-term covers Phase 1.*

1) **Group Name:** Individual project

2) **Team Name, Roll no:** Krishna Jhanwar, 12341250

3) **Novelty:**

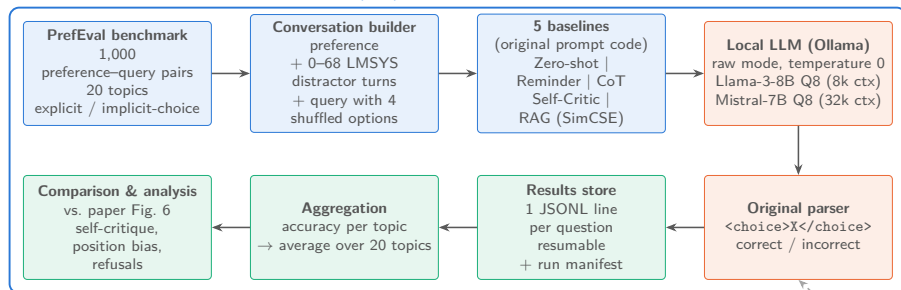
- **Overall idea:** a small *trained* classifier detects preference-stating turns and writes them to a structured memory (grows with the number of preferences, not turns), combined with a modern retriever (bge / e5, replacing 2021 SimCSE)
- **Phase 1 (done):** PrefEval's Bedrock-only evaluation rebuilt to run fully locally (Ollama, one 12 GB GPU) with the original prompts unchanged: resumable, logged, reproducible
- **Phase 1:** baselines reproduced on two of the paper's models (Llama-3-8B, Mistral-7B): 5 methods, explicit + implicit-choice, 0.2k–23k tokens, 64,535 questions; new implicit classification baselines
- **Phase 1:** found and fixed 3 prompt-formatting issues and 2 defects in the released retrieval data; new analyses (self-critique overturning answers, position bias at long context, RAG refusals)

4) **Contributions:**

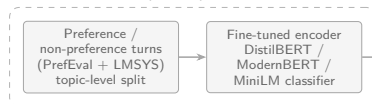
**Member 1** — Krishna Jhanwar (12341250): project design (all phases); Phase 1: local evaluation pipeline, debugging of prompt and data issues, all baseline runs ( $\approx 71$  GPU-hours), aggregation, analysis and reporting

# Architecture

## Phase 1 — evaluation harness & baseline reproduction (done)



## Phase 2 — classifier (planned)

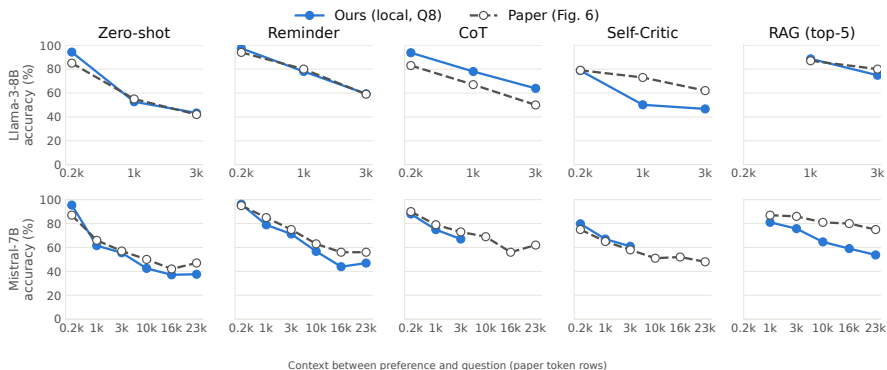


## Phase 3 — classifier + memory + retrieval (planned)



# Results and Findings in Tabular Format

Explicit preference: classification accuracy (%), 20 topics. Ours (solid) vs. paper Fig. 6 (dashed)



Explicit vs. implicit (choice) at 3k tokens (%)

| Method      | Llama-3-8B |          | Mistral-7B |          |
|-------------|------------|----------|------------|----------|
|             | Explicit   | Implicit | Explicit   | Implicit |
| Zero-shot   | 43.3       | 46.0     | 55.5       | 47.5     |
| Reminder    | 59.4       | 60.2     | 71.3       | 63.1     |
| CoT         | 63.9       | 61.7     | 67.0       | 59.5     |
| Self-Critic | 46.7       | 38.7     | 61.0       | 50.9     |
| RAG         | 74.8       | 73.7     | 75.7       | 64.0     |

Key findings

| Finding                         | Evidence                                        |
|---------------------------------|-------------------------------------------------|
| Baselines reproduced locally    | 12/29 cells within 5 pts; 20/29 within 10       |
| Accuracy decays with context    | Mistral Zero-shot 95.5 → 37.6 (0.2k → 23k)      |
| RAG most robust at long context | 23k: RAG 53.7 vs. Zero-shot 37.6                |
| Self-critique overturns answers | Llama, 0.2k: 205 right→wrong vs. 51 wrong→right |
| Long context → position prior   | 23k: accuracy 42 (answer A) vs. 23 (answer D)   |
| Implicit preferences harder     | Mistral: 7–15 pts below explicit                |
| Retrieval triggers refusals     | Llama RAG, 3k: 19.1% without a valid choice     |